



Slow Invertebrate Release Feeder

Eva Bochner, Rylan Carper, Sienna Giuseppi, Matthew Prato
Pratt School of Engineering, Duke University



Background



The North Carolina Zoo is the largest natural habitat zoo in the world.
Their Desert Dome exhibit is a glass-domed building designed to recreate an arid, colorful desert, home to free-flight birds and reptiles.

Motivation

Animals in captivity need environments and processes that mimic their natural habits to maintain their physical and mental well-being. However, the North Carolina Zoo's current feeding method in the Desert Dome exhibit—where insects are placed in a bowl—fails to replicate reptiles natural feeding behaviors.



Current Feeding Method:

- Reptiles feed too quickly
- Disrupts instinctual feeding habits
- Fosters competition between reptiles and birds

Project Goal

The team was tasked to create insect dispenser that randomly releases prey into the exhibit over 12 hours to mimic the unpredictability of the wild, encouraging reptiles to actively hunt. This more natural feeding method promotes better physical and mental well-being for the animals while enhancing their overall enrichment.

Design Criteria

Cricket Survival:	Insects leave the device alive
Recharge Frequency	Can run for one week
Maintenance	Easy for keepers to use and clean
Capacity	Holds 150 crickets
Durability	Withstands one year in the exhibit
Random Timing	Releases insects at random intervals
Cost	Costs less than \$100
Appearance	Blends into natural exhibit

Testing

Design Criteria	Test Description	Target Value	Results
Cricket Survival	Measure the internal temperature after 2 hours of running outside	Internal temperature less than 90°F	In Progress
Recharge Frequency	Run 5 trials of running the device for 2 hours, measuring voltage between trials	Battery still has charge after 10 hours	In Progress
Maintenance	User defined scale (1-5): Ask users to rate how easy it is to change the battery and refill the device	Scores greater than a 3.5 on user defined scale	PASSED- Scored an average of 4.9 on survey of 25 users.
Capacity	Measure volume of cricket compartments	Holds at least 150 crickets (1 cricket =0.1 cm ³)	PASSED- Dimensions (35.35 in ³) can hold 550+ crickets
Durability	Knocking over device 10 times	No damage to the device housing when struck	PASSED- No cracks in housing
Random Timing	Run for 2 hrs, record at what times each door drops	All 6 doors drop after different amounts of time	In Progress
Safety	Measure the distance between the ground and the released door	Doors do not swing below 6 in off the ground	PASSED- Door swings 8 inches above the ground.
Appearance	User defined scale (1-5): Ask users to rate how well the device blends into the exhibit	>3.5 on user defined scale	PASSED- Scored an average of 4.8 on survey of 25 users.

Current Solution

Structure

Use Case

- A) 9V battery:** Battery is rechargeable and can be removed from the device
- B) Lid:** Remove to fill device with crickets, mesh covering allows airflow for crickets
- C) Potentiometer:** By sliding the potentiometer, the user can choose the length of time in which the dispenser runs within a range of 1 to 12 hours
- D) On/off switch:** Activates device
- E) Arduino:** Code choses a random amount of time within the time value set by the potentiometer and sends signals to pins one at a time
- F) 6 relays:** Amplifies the voltage supplied by Arduino pin from 5V to 12V
- G) 6 solenoids:** When voltage is received, the solenoid retracts
- H) 6 trap doors:** Retracting solenoid drops door, releasing contents
- I) Partitions:** Partition can be removed for easy containment cleaning



Figure 1. Isometric view



Figure 2. Side view



Figure 3. Bottom view



Figure 4. Top view



Figure 5. Lid

Tan color and sand texture help to camouflage into the dry desert exhibit

Acrylic 'T' lid keeps electronics accessible for maintenance, yet safe from external factors

Conclusion

Our team successfully created an electronic feeder for the reptiles in the Desert Dome Exhibit that dispenses crickets at random time intervals over a period from 1-12 hours, implementing natural feeding habits back into the lives of the reptiles in the Dome

Future Work

To improve this solution, we would like to investigate different solenoids that can function with 5V instead of 12V, which would allow us to remove the relays. Less wired components would make the device more efficient for maintenance. Additionally, we would like to further our design by incorporating a mechanism to dispense vitamin powder supplement onto the crickets before release, as this provides more nutrition to the reptiles and is currently done manually by keepers

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